#1

Solution. For 1), for any compact set K in $\mathbb{R}^n \times \mathbb{R}$, one can find a cover $B_R \times [-T, T]$ such that $K \subset B_R \times [-T, T]$. ($T, R$ are finite) Hence

$$\int_K H_e \leq \int_0^T \int_{B_R} H_e dx dt \leq \int_0^T dt = T < \infty$$

To show it is a distribution, refer to Example 1.27 in suppnotes.
For 2), Recall here the convergence are determined by weak* topology in dual space, first show that $H_\eta(\phi) \to H_e(\phi)$ for every test function $\phi \in C_c^\infty(\mathbb{R}^n \times \mathbb{R})$.

$$|H_\eta(\phi) - H_e(\phi)| = |\int_K (H_\eta - H_e)(x,t)\phi(x,t))| \leq \int_0^\eta \int_{\mathbb{R}^n} |H(x,t)\phi(x,t)| dx dt$$

As $\int H(x,t)dx$ equal to some constant, H nonegative. Next, ϕ is bounded, turn $\eta \to 0$, we have $|H_\eta(\phi) - H_e(\phi)| \to 0$ for any test function ϕ. Also note that such ϕ_e as a limit is unique ($\mathcal{D}'(U)$ is Hausdorff with above weak star topology).

To show $(\partial_t - \Delta)H_\eta \to \delta$, $H_\eta = H \cdot \mathbf{1}_{\{t>\eta\}}$, then

$$\partial_t H_\eta = \partial_t H \cdot \mathbf{1}_{\{t>\eta\}} + H \cdot \delta_\eta$$

So that for

$$< (\partial_t - \Delta)H_\eta, \phi >= \int H(x,\eta)\phi(x,\eta)dx \to \phi(0,0)$$

by approxiamte to identity and $(\partial_t - \Delta)H = 0$ on $t > \eta$.

For the last step, by the property of Heat kernel, we can exchange the differetial sign with integral thus proof $(\partial_t - \Delta)H_e = \delta$ ■

#4

Solution. For a), one just still consider Fourier Transform, $\mathcal{F}$. Then we have

$$\partial_t \hat{u} + i|\xi|^2 \hat{u} = 0 \quad \hat{u}|_{t=0} = \hat{g}$$

Solve Above ODE, one get $\hat{u}(\xi, t) = e^{-it|\xi|^2}\hat{g}(\xi)$. Change variable, we obtain,

$$u(t,x) = (4\pi i t)^{-n/2} \int e^{-|x-y|^2/4it} g(y) dy$$

All above operation is legal as $e^{-|x|^2}, g \in \mathcal{S}$.
For b), as $|e^{-|x-y|^2/4it}| = |e^{i|x-y|^2/4t}| = 1$, then it is a trivial result.
For c),

$$E'(t) = \int \partial_t |u(t,x)|^2 dx = -2\mathrm{Re} \int |\nabla u|^2 dx \leq 0$$

and $E(t) \geq 0$ on $t > 0$, then $E = |g|_{L^2}$ as a constant on $\mathbb{R}_+$.
For d), by plancherel equality,

$$|\hat{u}|_{L^2} = |u|_{L^2} = |\hat{g}|_{L^2} = |g|_{L^2} = E(0)$$

■